ANALYSIS OF ADVERSE EVENTS, ROBOTIC UROLOGICAL SURGERIES

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INTRODUCTION

Background:

- Robotic instruments are commonplace in ORs, clinics, and most healthcare settings. Within surgical fields like urology, surgeries are often robot-assisted (RA). Examples include procedures like the RA radical prostatectomy (RARP), RA simple prostatectomy (RASP), RA radical cystectomy (RC), RA nephroureterectomy (RANU), and RA partial nephrectomy (RAPN).
- Compared to laparoscopic alternatives, RA surgeries provide 3D view (vs 2D), greater dexterity via robotic arms, smaller incisions, shorter hospital stays (cheaper), enhanced visualization, reduced physician fatigue, etc.
- Despite these advantages, device-related adverse events remain critical area of study. Thus, the MAUDE (Manufacturer and User Facility Device Experience) database, maintained by the FDA, is a valuable resource for quantifying the potential harm, or lack thereof, of these devices.

Objective and Significance:

- Using data from the MAUDE database, we aim to:
  - **(1)** Systematically analyze/compare the performance and adverse event profiles of the SynchroSeal and Vessel Sealer Extend devices (SS and VSE, respectively) during urologic procedures. These devices are use to seal and cut blood vessels.
  - **(2)** Provide insight which enhances device design, informs future research, and improves patient outcomes.
- Finally, on March 3rd, 2022, the FDA issued a field safety notice about both of these devices, cautioning customers about issues sealing when excessive tissue is in the instrument jaws of the VSE or SS instruments. This problem may have contributed to many adverse events, necessitating this research.

METHODS

MAUDE query and sample

- MAUDE was queried for the following terms
  - *Procedures*: prostatectomy, nephrectomy, cystectomy, lymphadenectomy, nephroureterectomy, pyeloplasty
  - *Devices:* SynchroSeal and Vessel Sealer Extend
  - *Note:* Queries were restricted to urological reports due to time constraints.
- N = 37 and 30 for SS and VSE, respectively.

Classification and Analyses:

- Adverse events were classified according to severity and the relationship of the event to the device being used as outlined in Table 1 and Table 2.
- Similarities between events were subsequently grouped and compared for each device.
- Chi square analyses were used to compare the severity of the event profiles of each device. This analysis was repeated after removing "unrelated" events.

Table 1: Severity Classification

Severity of Event		Classification
MAUDE 1	"Mild"	(a) **NO** harm occurring to patient or staff ***and*** (b) **NO** significant deviation from planned procedure
MAUDE 2	"Moderate"	(a) Harm to patient or staft requiring **minor** intervention ***and/or*** (b) Some deviation from planned procedure without change in surgical outcome
MAUDE 3	"Severe"	(a) Harm to patient requiring **major** intervention ***and/or*** (b1) requiring **significant deviation** from the planned procedure intra-operatively ***or*** (b2) requiring **aggressive intervention**
MAUDE 4	"Critical"	(a) **Life-threatening** event ***or*** (b) **Death** during procedure

Table 2: Device Relationship Classification

Relationship of event to device		Events include
Level 1	"Unrelated"	**Clearly unrelated** event, adverse event related to patient's baseline health, or surgeon's or user's error; not related to the synchroseal or vessel sealer extend
Level 2	"Possibly Related"	Event may be related to device, but some **ambiguity remains** (i.e. may be due to another device used in conjunction).
Level 3	"Definitely Related"	**Event is clearly related to device malfunctioning** (the synchroseal or vessel sealer extend); examples include system error, causing case to be aborted or device breaking, causing need to switch instruments or retrieve parts.

Table 3: Freq. of adverse events by severity and devices

	SS n = 37 n (%)	VSE n = 30 n (%)
Mild	17 (46)	20 (67)
Moderate	17 (46)	5 (17)
Severe	1 (2.7)	3 (10)
Critical	2 (5.4)	2 (6.7)
Sum	37 (100)	30 (100)

Table 4: Freq. of adverse events by relationship to devices

	SS n = 37 n (%)	VSE n = 30 n (%)
Unrelated	4 (11)	2 (7)
Possibly Related	2 (5)	18 (60)
Definitely Related	31 (84)	10 (33)
Sum	37 (100)	30 (100)

CONCLUSION

Event severity and Frequency

- The SS device exhibited a higher frequency of device-related events compared to the VSE, with significant differences in the severity of events after removing events unrelated to device usage *(p = 0.025)*.

Specific Defects:

- Grouping of adverse events revealed that device detachment and thermal damage were more prevalent in the SS device, suggesting potential areas for design improvement.
- The most common issues with the SS include dislodged device components, jaw cover tears, and thermal damage.
- The VSE event reports are still being analyzed and grouped.

Impact of FDA Safety Notice:

- The sealing/cutting issues described in these event reports further emphasizes the importance of adhering to safety guidelines and the need for ongoing research of robotic surgical devices.

Recommendations for Device Improvement:

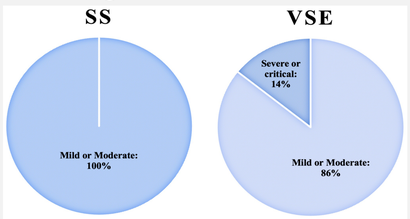
- This work and similar future studies may help refine the mechanical components of sealing instruments to improve common issues like device component dislodgement, thermal regulation, and coagulation issues.

Takeaway:

- This study provides insight that can inform future research, guide clinical practice, and enhance the safety and effectiveness of robotic-assisted urological surgeries.

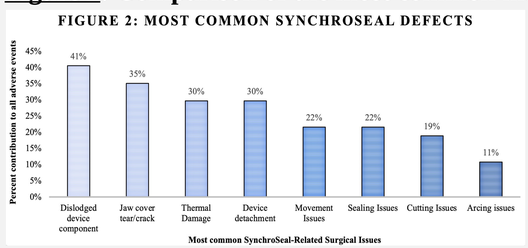
RESULTS

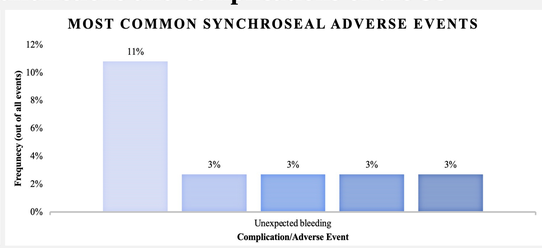
Figure 1: Grouped severity comparison after removing unrelated events



Compared to the VSE, the SS had significantly different (less) severe or critical events after removing events which were unrelated to device usage. **(p = 0.025)**

Figure 2: Comparison of the most common malfunctions and complications of the SS





FUTURE

Moving Forward:

- Future studies utilizing the MAUDE database should focus on non-urological data to gain greater power, further assess the safety of these devices, and explore modifications which could improve patient outcomes.

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SOURCES